

YEAR 7 MATHEMATICS

Chapter 7

Decimals

MIXED PRACTICE WORKSHEETS

7

How to use this booklet

- Each lesson is **one double-sided sheet**. Write your name, class and date at the top.
- Then work down the **columns**, left to right. They get harder as you go: **Easy** **Easy** **Medium** on page 1, then **Medium** **Challenging** **Challenging** on page 2.
- Most questions practise **today's skill**. A question marked **from 1.03** comes from an earlier lesson, to keep it fresh. Stuck? Look back at that lesson.
- Each question tells you what to do: read it carefully. Show your working on the lines. The first question with working in each Easy column shows you how to set it out.

WE DO

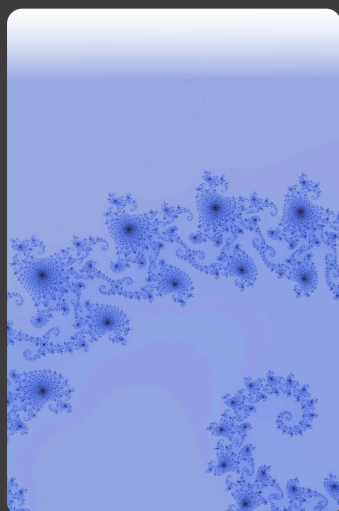
The grey box at the top. Your teacher shows the key idea with two or three examples. **Copy every step** into the space.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
7.01	Ordering decimals	3		☐
7.02	Decimals and fractions	5		☐
7.03	Adding and subtracting decimals	7		☐
7.04	Multiplying and dividing by powers of 10	9		☐
7.05	Using a known product	11		☐
7.06	Multiplying decimals	13		☐
7.07	Dividing decimals by whole numbers	15		☐
7.08	Dividing by a decimal	17		☐
7.09	Fractions to decimals	19		☐
7.10	Rounding decimals	21		☐
7.11	Decimals and money	23		☐

In this chapter you will

- order decimals
- convert between decimals and fractions
- add, subtract, multiply and divide decimals
- multiply and divide by powers of 10
- round decimals and write recurring decimals
- solve money problems



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is a double spiral deep inside Seahorse Valley. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

Compare decimals place by place from the left. Give them the same number of decimal places first: $0.5 = 0.50$, so 0.5 is larger than 0.45 .

Watch your teacher.
Copy every step.

E1 Which is larger: 0.7 or 0.65 ?

E2 Order from smallest to largest: 2.3 , 2.03 , 2.33 , 2.303

E3 Write $<$ or $>$: 4.09 $\square$ 4.1

Easy

Questions 1–9

1 Which is larger: 0.4 or 0.79 ?

2 What is the value of the 1 in 0.118 ?

3 Write $<$ or $>$: 6.56 $\square$ 6.4

4 Which is larger: 0.7 or 0.79 ?

5 What is the value of the 1 in 4.153 ?

6 Write $<$ or $>$: 6.01 $\square$ 6.4

7 Which is larger: 0.1 or 0.76 ?

8 What is the value of the 3 in 7.763 ?

9 Write an expression: 6 less than b

from 5.04

Easy

Questions 10–18

10 What is the value of the 1 in 3.171 ?

11 Write $<$ or $>$: 3.29 $\square$ 3.5

12 Which is larger: 0.1 or 0.18 ?

13 Write 8 out of 15 as a fraction in simplest form.

from 4.12

14 Order from smallest to largest: 2.03 , 2.33 , 2.3 , 2.303

15 What is the value of the 4 in 7.943 ?

16 Write $<$ or $>$: 6.76 $\square$ 6.9

17 What is 10% of 772 ?

from 4.11

18 Which is larger: 0.7 or 0.4 ?

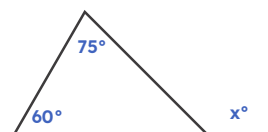
Medium

Questions 19–26

19 Order from smallest to largest: 9.707 , 9.07 , 9.7 , 9.77

20 Order from largest to smallest: 0.04 , 0.5 , 0.94 , 0.762

21 Find x , the exterior angle.



from 6.07

22 Write a decimal between 0.4 and 0.5 .

23 Order from smallest to largest: 7.04 , 7.404 , 7.44 , 7.4

24 Write $8bt$ in expanded form.

from 5.03

25 Order from largest to smallest: 0.594 , 0.01 , 0.1 , 0.62

26 Write a decimal between 0.2 and 0.3 .

Medium

Questions 27–36

- 27 Order from largest to smallest: 0.07, 0.17, 0.8, 0.795

- 28 Write a decimal between 0.7 and 0.8.

- 29 How many axes of symmetry does the capital letter S have?

from 6.03

- 30 Order from smallest to largest: 9.05, 9.505, 9.55, 9.5

- 31 A rectangle is 4m cm long and 9y cm wide. Write its area.

from 5.08

- 32 Which is closest to 2: 1.9, 2.01, 2.1, 1.99?

- 33 Order from largest to smallest: 0.15, 0.495, 0.09, 0.3

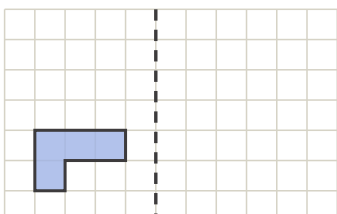
- 34 Solve in your head: $\frac{x}{2} = 10$

from 5.10

- 35 Write a decimal between 0.1 and 0.2.

- 36 Reflect the shape in the dashed line. Draw the image.

from 6.01



Challenging

Questions 37–46

- 37 Which is closest to 2: 2.01, 2.1, 1.99, 1.9?

- 38 Jack says $0.35 > 0.3$ because $35 > 3$. Is that right? Explain.

- 39 A number plus 2 is 21. What is the number?

from 5.13

- 40 Order from smallest to largest: -0.7 , 0.3 , 0 , -0.9

- 41 $41 + 33 + 59 + 29$

from 5.01

- 42 Which is closest to 5: 5.1, 5.04, 4.95, 4.9?

- 43 Jack says $0.38 > 0.4$ because $38 > 4$. Is that right? Explain.

- 44 Classify a triangle with sides 8 cm, 8 cm, 8 cm by its sides.

from 6.05

- 45 Order from smallest to largest: -0.36 , -0.3 , 0 , 0.2

- 46 $P = 2l + 2w$. Find P when $l = 9$ and $w = 10$.

from 5.05

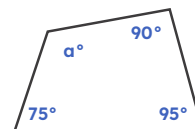
Challenging

Questions 47–56

- 47 Order from smallest to largest: -0.39 , 0 , 0.6 , -0.9

- 48 Which is closest to 2: 1.9, 2.1, 1.99, 2.01?

- 49 Find a . (Not to scale.)

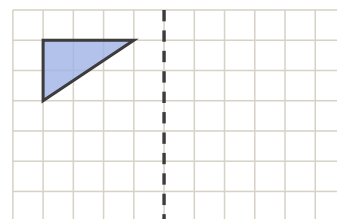


from 6.09

- 50 Write a decimal between 0.6 and 0.7.

- 51 Reflect the shape in the dashed line, then translate the image 2 down. Draw both images.

from 6.02



- 52 Order from smallest to largest: -0.88 , 0 , -0.1 , 0.5

- 53 Which is closest to 1: 1.09, 1.1, 0.99, 0.9?

- 54 Simplify $4a - 3a$.

from 5.07

- 55 Write a decimal between 0.5 and 0.6.

- 56 A shape has rotational symmetry of order 5. What is its smallest angle of turn?

from 6.04

NAME _____ CLASS _____ DATE _____

WE DO
KEY IDEA

After the point come tenths, hundredths and thousandths: $0.37 = 37$ hundredths = $\frac{37}{100}$. To write a fraction as a decimal, make its denominator 10, 100 or 1000.

Watch your teacher.
Copy every step.

E1 Write 0.45 as a fraction in simplest form.

E2 Write $\frac{3}{5}$ as a decimal.

E3 Write 2.125 as a mixed numeral.

Easy

Questions 1–9

1 Write 0.1 as a fraction.

2 Write $\frac{28}{100}$ as a decimal.

3 Write $\frac{378}{1000}$ as a decimal.

4 Write 0.4 as a fraction.

5 Write $\frac{15}{100}$ as a decimal.

6 Write $\frac{718}{1000}$ as a decimal.

7 Write 0.2 as a fraction.

8 Write $\frac{14}{100}$ as a decimal.

9 What is the value of the 9 in 9.149?

from 7.01

Easy

Questions 10–18

10 Write $\frac{99}{100}$ as a decimal.

11 Write $\frac{472}{1000}$ as a decimal.

12 Write 0.7 as a fraction.

13 Write an expression: 9 less than m

from 5.04

14 Write 0.54 as a fraction in simplest form.

$$= \frac{54}{100}$$

$$= \frac{27}{50}$$

15 Write $\frac{48}{100}$ as a decimal.

16 Write $\frac{24}{1000}$ as a decimal.

17 Write $1\frac{1}{2}$ as an improper fraction.

from 4.01

18 Write 0.3 as a fraction.

Medium

Questions 19–26

19 Write 0.65 as a fraction in simplest form.

20 Write $\frac{3}{20}$ as a decimal.

21 Write a decimal between 0.3 and 0.4.

from 7.01

22 Write 8.71 as a mixed numeral.

23 Write 0.72 as a fraction in simplest form.

24 How many axes of symmetry does the capital letter M have?

from 6.03

25 Write $\frac{3}{4}$ as a decimal.

26 Write 1.58 as a mixed numeral.

Medium

Questions 27–37

- 27 Write $\frac{2}{5}$ as a decimal.

- 28 Write 2.45 as a mixed numeral.

- 29 Write $\frac{10}{6}$ as a mixed numeral.

from 4.01

- 30 Write 0.02 as a fraction in simplest form.

- 31 Order from smallest to largest: 7.06, 7.66, 7.606, 7.6

from 7.01

- 32 Write 0.375 as a fraction in simplest form.

- 33 Write $\frac{1}{4}$ as a decimal.

- 34 $19 + 15 + 81 + 79$

from 5.01

- 35 Write 8.46 as a mixed numeral.

- 36 Simplify $\frac{25}{30}$.

from 4.01

- 37 Write 0.78 as a fraction in simplest form.

Challenging

Questions 38–47

- 38 Write 0.875 as a fraction in simplest form.

- 39 Order from smallest to largest: 0.48, $\frac{1}{2}$, $\frac{3}{5}$

- 40 Order from largest to smallest: 0.9, 0.833, 0.43, 0.02

from 7.01

- 41 Priya says 0.3 is the same as $\frac{1}{3}$. Is that right? Explain.

- 42 Find k^2 when $k = 11$.

from 5.05

- 43 Write 0.025 as a fraction in simplest form.

- 44 Order from smallest to largest: $\frac{4}{5}$, 0.7, $\frac{3}{4}$

- 45 Write $2\frac{1}{4}$ as an improper fraction.

from 4.01

- 46 Write 0.125 as a fraction in simplest form.

- 47 Write a decimal between 0.2 and 0.3.

from 7.01

Challenging

Questions 48–56

- 48 Order from smallest to largest: $\frac{3}{4}$, $\frac{4}{5}$, 0.7

- 49 Write 0.625 as a fraction in simplest form.

- 50 A triangle has angles 95° , 30° and 55° . Classify it by its angles and by its sides.

from 6.05

- 51 Write 6.36 as a mixed numeral.

- 52 Write $\frac{5}{4}$ as a mixed numeral.

from 4.01

- 53 Order from smallest to largest: $\frac{4}{5}$, $\frac{3}{4}$, 0.7

- 54 Write 0.064 as a fraction in simplest form.

- 55 Order from smallest to largest: 2.01, 2.101, 2.1, 2.11

from 7.01

- 56 Write 3.01 as a mixed numeral.

NAME _____ CLASS _____ DATE _____

WE DO

Watch your teacher.
Copy every step.

KEY IDEA

Line up the decimal points. Fill any gaps with zeros so both numbers have the same number of decimal places, then add or subtract as with whole numbers.

E1 $3.45 + 2.8$

E2 $7 - 2.36$

E3 A \$20 note pays for a \$13.85 meal. Find the change.

Easy

Questions 1–16

1 $9.6 + 7.8$

2 $8.6 - 4.7$

3 $6.68 + 9.17$

4 $9.4 + 9.7$

5 $5.6 - 4.6$

6 $5.29 + 8.68$

7 $4.4 + 2.4$

8 $9.1 - 3.8$

9 What is the value of the 1 in 8.711?

from 7.01

10 $3.99 + 4.72$

11 $1.8 + 3$

12 $5.2 - 3.2$

13 $4.58 + 6.67$

14 $2.8 + 9.3$

15 $9.7 - 1.2$

16 $5.08 + 6.37$

Easy

Questions 17–27

17 $5.2 - 3.4$

18 $3.09 + 5.15$

19 $2.4 + 3.2$

20 Work out $4 \times (20 + 7)$ two ways.

from 5.02

21 $3.35 + 1.8$

= ____

22 $8.3 - 3$

23 $2.31 + 4.96$

24 Write $\frac{2}{100}$ as a decimal.

from 7.02

25 $2.1 + 7.9$

26 $2.53 + 3.3$

27 $5.8 - 3.8$

Medium

Questions 28–35

28 $6.58 + 6.9$

29 $14 - 2.28$

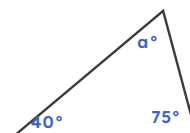
30 Write a decimal between 0.8 and 0.9.

from 7.01

31 A \$20 note pays for a \$2.07 meal. Find the change.

32 $2.82 + 7.3$

33 Find α .



from 6.06

34 $8 - 2.88$

35 A \$50 note pays for a \$2.12 meal. Find the change.

Medium

Questions 36–46

36 $10 - 2.02$

37 A \$20 note pays for a \$3.06 meal. Find the change.

 38 Write $\frac{195}{1000}$ as a decimal.

from 7.02

39 $2.59 + 1.3$

40 Order from smallest to largest: 8.06, 8.606, 8.66, 8.6

from 7.01

41 $6 + 2.3 + 8$

42 $4 - 2.07$

43 $34 + 30 + 66 + 37$

from 5.01

44 A \$50 note pays for a \$15.16 meal. Find the change.

45 Write 0.4 as a fraction.

from 7.02

46 $8.07 + 2.4$

Challenging

Questions 47–56

47 $3.2 + 8.15 + 3$

48 Aisha runs 4.56 km on Monday and 4.97 km on Tuesday. How much further is that than 5 km?

49 Order from largest to smallest: 0.865, 0.7, 0.86, 0.01

from 7.01

 50 Find the missing number:
 $8.31 + \square = 17.38$

51 Solve $4n + 8 = 20$.

from 5.12

 52 Noah adds $7.3 + 5.88$ by lining up the last digits and gets 6.61. What went wrong? What is the right answer?

53 $4.7 + 8.58 + 3$

 54 Write $\frac{94}{100}$ as a decimal.

from 7.02

55 Lena runs 2.55 km on Monday and 4.18 km on Tuesday. How much further is that than 5 km?

56 Write a decimal between 0.1 and 0.2.

from 7.01

Challenging

Questions 57–66

57 Chloe runs 3.93 km on Monday and 4.11 km on Tuesday. How much further is that than 5 km?

 58 Find the missing number:
 $2.33 + \square = 10.54$

59 11 equal packs hold 231 pencils. How many pencils are in each pack?

from 5.13

 60 Ruby adds $7.8 + 8.6$ by lining up the last digits and gets 9.38. What went wrong? What is the right answer?

 61 Write $\frac{724}{1000}$ as a decimal.

from 7.02

62 $2.9 + 6.12 + 1$

63 A \$50 note pays for a \$17.20 meal. Find the change.

64 Order from smallest to largest: 7.909, 7.99, 7.9, 7.09

from 7.01

65 Zac runs 4.51 km on Monday and 2.01 km on Tuesday. How much further is that than 5 km?

 66 Write an expression: 6 less than b

from 5.04

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

To multiply by 10, 100 or 1000, move every digit 1, 2 or 3 places to the left (the decimal point seems to move right). To divide, move the digits to the right.

Watch your teacher.
Copy every step.

E1 3.47×100

E2 $52.6 \div 1000$

E3 Change 4.5 kg to grams.

Easy

Questions 1–16

1 $6.91 \times 1\,000$ 2 $96.4 \div 10$

3 $14 \div 100$ 4 8.92×10

5 $70.2 \div 1\,000$ 6 $6 \div 10$

7 4.88×10 8 $2.3 \div 10$

9 $7.8 - 2.7$

from 7.03

10 $21 \div 10$ 11 5.73×100

12 $89.8 \div 1\,000$ 13 $15 \div 100$

14 1.31×10 15 $42 \div 10$

16 $37 \div 100$

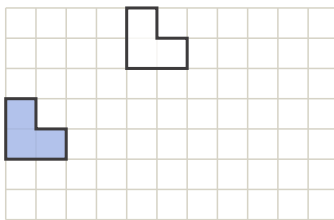
Easy

Questions 17–27

17 $66.1 \div 10$ 18 $21 \div 100$

19 7.68×10

20 Describe the translation from the shaded shape to the white shape.



from 6.01

21 Change 9.7 km to m.

___ km = ___ m

$9.7 \times 1\,000 =$ _____

22 $90.1 \div 1\,000$ 23 $28 \div 100$

24 68×10

from 3.02

25 8.74×100

26 Change 0.34 km to m.

27 $60.5 \div 1\,000$

Medium

Questions 28–35

28 Change 8.97 kg to g.

29 Change 32 cm to m.

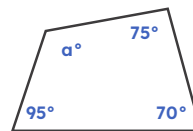
30 Write $\frac{31}{50}$ as a decimal.

from 7.02

31 Find the missing number:
 $2.6 \times \square = 260$

32 Change 2.83 L to mL.

33 Find a . (Not to scale.)



from 6.09

34 Change 630 mm to cm.

35 Find the missing number:
 $8.97 \times \square = 897$

Medium

Questions 36–46

- 36 Change 210 mm to cm.

- 37 Find the missing number:
 $1.71 \times \square = 1\,710$

- 38 What is the value of the 7 in 6.167?

from 7.01

- 39 Change 3.08 cm to mm.

- 40 A \$20 note pays for a \$2.06 meal. Find the change.

from 7.03

- 41 Find the missing number:
 $\square \times 1\,000 = 95$

- 42 Change 254 mm to cm.

- 43 Simplify $\frac{18x}{6}$.

from 5.09

- 44 Find the missing number:
 $9.06 \times \square = 90.6$

- 45 70×8

from 3.02

- 46 Change 8.75 cm to mm.

Challenging

Questions 47–56

- 47 Find the missing number:
 $\square \times 100 = 3.3$

- 48 $42.5 \times 100 \div 1000$

- 49 Write 4.26 as a mixed numeral.

from 7.02

- 50 One pen costs \$1.50. How much do 1000 pens cost?
How much do 10 pens cost?

- 51 Write an expression: 7 less than m

from 5.04

- 52 Find the missing number:
 $\square \times 1\,000 = 79$

- 53 $37.8 \times 100 \div 1000$

- 54 Write $<$ or $>$: $8.24 \square 8.7$

from 7.01

- 55 One pen costs \$5.37. How much do 1000 pens cost?
How much do 10 pens cost?

- 56 $1.18 + 7.3$

from 7.03

Challenging

Questions 57–66

- 57 $94.4 \times 100 \div 1000$

- 58 One pen costs \$8.11. How much do 1000 pens cost?
How much do 10 pens cost?

- 59 A rectangle is $5c$ long and $4t$ wide. Write its perimeter as simply as possible.

from 5.07

- 60 Find the missing number:
 $\square \times 1\,000 = 89$

- 61 44×6

from 3.02

- 62 Find the missing number:
 $1.26 \times \square = 12.6$

- 63 $69.8 \times 100 \div 1000$

- 64 Write 0.56 as a fraction in simplest form.

from 7.02

- 65 One pen costs \$6.84. How much do 1000 pens cost?
How much do 10 pens cost?

- 66 A shape is turned 180° clockwise, then 180° clockwise. What single turn does the same?

from 6.02

NAME _____ CLASS _____ DATE _____

WE DO
KEY IDEA

Use a product you know. If $23 \times 4 = 92$, then $2.3 \times 4 = 9.2$ and $0.23 \times 4 = 0.92$. Count the decimal places in the question: the answer has the same number.

Watch your teacher.
Copy every step.

E1 Given $36 \times 7 = 252$, find 3.6×7 .

E2 Given $36 \times 7 = 252$, find 0.36×0.7 .

E3 Given $252 \div 7 = 36$, find $25.2 \div 7$.

Easy

Questions 1–9

1 Given $31 \times 9 = 279$, find 3.1×9 .

2 Given $15 \times 9 = 135$, find 15×0.9 .

3 Given $21 \times 3 = 63$, find 0.21×3 .

4 Given $39 \times 6 = 234$, find 3.9×6 .

5 Given $35 \times 3 = 105$, find 35×0.3 .

6 Given $40 \times 3 = 120$, find 0.4×3 .

7 Given $15 \times 9 = 135$, find 1.5×9 .

8 Given $41 \times 6 = 246$, find 41×0.6 .

9 $17.6 \div 100$

from 7.04

Easy

Questions 10–18

10 Given $15 \times 5 = 75$, find 15×0.5 .

11 Given $30 \times 8 = 240$, find 0.3×8 .

12 Given $39 \times 8 = 312$, find 3.9×8 .

13 Classify a triangle with sides 11 cm, 11 cm, 11 cm by its sides.

from 6.05

14 Given $37 \times 3 = 111$, find 3.7×0.3 .

____ + ____ = ____ decimal places

15 Given $22 \times 6 = 132$, find 22×0.6 .

16 Given $38 \times 9 = 342$, find 0.38×9 .

17 $6.5 - 4.2$

from 7.03

18 Given $31 \times 5 = 155$, find 3.1×5 .

Medium

Questions 19–26

19 Given $14 \times 7 = 98$, find 1.4×0.7 .

20 Given $215 \div 5 = 43$, find $21.5 \div 5$.

21 Order from largest to smallest: 0.391, 0.85, 0.9, 0.05

from 7.01

22 Given $259 \times 7 = 1\,813$, find 2.59×7 .

23 Given $36 \times 8 = 288$, find 3.6×0.8 .

24 Write $\frac{3}{25}$ as a decimal.

from 7.02

25 Given $288 \div 8 = 36$, find $28.8 \div 8$.

26 Given $325 \times 5 = 1\,625$, find 3.25×5 .

Medium

Questions 27–37

- 27 Given $102 \div 3 = 34$, find $10.2 \div 3$.

- 28 Given $370 \times 6 = 2\,220$, find 3.7×6 .

- 29 Find k^2 when $k = 7$.

from 5.05

- 30 Given $24 \times 7 = 168$, find 2.4×0.7 .

- 31 Write $<$ or $>$: $6.1 \square 6.3$

from 7.01

- 32 Given $39 \times 26 = 1\,014$, find 3.9×0.26 .

- 33 Given $258 \div 6 = 43$, find $25.8 \div 6$.

- 34 $48 \div 10$

from 7.04

- 35 Given $173 \times 3 = 519$, find 1.73×3 .

- 36 A \$50 note pays for a \$1.56 meal. Find the change.

from 7.03

- 37 Given $20 \times 3 = 60$, find 2×0.3 .

Challenging

Questions 38–47

- 38 Given $18 \times 27 = 486$, find 1.8×0.27 .

- 39 Given $12 \times 13 = 156$, find 120×1.3 .

- 40 Write an expression: 11 more than 4 times t

from 5.04

- 41 Given $16 \times 5 = 80$, find $0.8 \div 0.5$.

- 42 Write $\frac{428}{1000}$ as a decimal.

from 7.02

- 43 Given $34 \times 25 = 850$, find 3.4×0.25 .

- 44 Given $23 \times 19 = 437$, find 230×1.9 .

- 45 Order from smallest to largest: 2.9, 2.99, 2.09, 2.909

from 7.01

- 46 Given $40 \times 6 = 240$, find $2.4 \div 0.6$.

- 47 Change 5.53 m to cm.

from 7.04

Challenging

Questions 48–57

- 48 Given $25 \times 22 = 550$, find 250×2.2 .

- 49 Given $20 \times 4 = 80$, find $0.8 \div 0.4$.

- 50 Simplify $n \times n$.

from 5.03

- 51 Given $14 \times 24 = 336$, find 1.4×0.24 .

- 52 What is the value of the 3 in 7.932?

from 7.01

- 53 Given $440 \times 8 = 3\,520$, find 4.4×8 .

- 54 Given $16 \times 23 = 368$, find 160×2.3 .

- 55 $8.9 + 7.9$

from 7.03

- 56 Given $31 \times 4 = 124$, find $1.24 \div 0.4$.

- 57 Write 0.34 as a fraction in simplest form.

from 7.02

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

Multiply as if there were no decimal points. Then count the decimal places in both numbers: the answer has that many. Estimate to check.

Watch your teacher.
Copy every step.

E1 0.3×0.4

E2 2.5×1.2

E3 4.6×0.3

Easy

Questions 1–16

1 0.1×5

2 0.8×0.5

3 6.4×6

4 0.9×6

5 0.2×0.6

6 6.1×2

7 0.9×4

8 0.1×0.3

9 Given $31 \times 3 = 93$, find 31×0.3 .

from 7.05

10 2.5×3

11 0.1×9

12 0.5×0.7

13 4.8×2

14 0.8×8

15 0.9×0.5

16 7.8×8

Easy

Questions 17–27

17 0.1×0.6

18 7.7×3

19 0.7×7

20 What is 10% of 621?

from 4.11

21 2.6×0.3

$26 \times 3 =$ _____

$=$ _____

22 0.9×0.9

23 6.9×3

24 $33.9 \div 10$

from 7.04

25 0.5×5

26 7.1×0.5

27 0.2×0.8

Medium

Questions 28–35

28 3.1×0.7

29 7.93×4

30 Order from largest to smallest: 0.494, 0.5, 0.21, 0.04

from 7.01

31 A rectangle is 1.1 m by 1.8 m. Find its area.

32 $(0.8)^2$

33 $17 - 1.38$

from 7.03

34 9.2×0.4

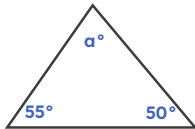
35 4.56×4

Medium

Questions 36–46

36 1.48×9

- 37 A rectangle is 4.7 m by 1.5 m.
-
- Find its area.

38 Find a .

from 6.06

39 $(0.4)^2$

40 Write $\frac{61}{100}$ as a decimal.

from 7.02

41 4×0.9

- 42 Apples cost \$5.70 per kg. Find the cost of 4.8 kg, to the nearest cent.

43 Given $42 \times 6 = 252$, find 0.42×6 .

from 7.05

44 5.52×3

45 Find the missing number:
 $6.45 \times \square = 64.5$

from 7.04

- 46 A rectangle is 2.4 m by 1.1 m.
-
- Find its area.

Challenging

Questions 47–56

- 47 Apples cost \$2.09 per kg. Find the cost of 1.4 kg, to the nearest cent.

48 7.5×0.73

- 49 A shape has rotational symmetry of order 3. What is its smallest angle of turn?

from 6.04

- 50 Is
- 0.8×0.9
- larger or smaller than 0.8? Explain.

51 $7.52 + 3.03$

from 7.03

- 52 Apples cost \$8.07 per kg. Find the cost of 3.6 kg, to the nearest cent.

53 1.2×0.14

- 54 Write a decimal between 0.1 and 0.2.

from 7.01

- 55 Is
- 2.5×0.4
- larger or smaller than 2.5? Explain.

56 Given $34 \times 5 = 170$, find 3.4×0.5 .

from 7.05

Challenging

Questions 57–66

57 1.6×0.45

- 58 Is
- 0.6×0.3
- larger or smaller than 0.6? Explain.

- 59 A rhombus has a side of 10 cm. Find its perimeter.

from 6.10

- 60 Apples cost \$5.72 per kg. Find the cost of 3.3 kg, to the nearest cent.

61 Write $\frac{876}{1000}$ as a decimal.

from 7.02

62 $(0.7)^2$

63 2.9×0.27

64 3.28×100

from 7.04

- 65 Is
- 0.4×0.5
- larger or smaller than 0.4? Explain.

66 $2.99 + 3$

from 7.03

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

Divide as usual and keep the decimal point in the same place in the answer. Write extra zeros after the decimal point if you need to keep dividing.

Watch your teacher.
Copy every step.

E1 $8.4 \div 4$

E2 $7.5 \div 6$

E3 \$14.70 is shared by 3 people. How much each?

Easy

Questions 1–16

1 $5.4 \div 3$

2 $47.88 \div 7$

3 $1.4 \div 2$

4 $36 \div 8$

5 $12.35 \div 5$

6 $1 \div 5$

7 $46.9 \div 7$

8 $68.08 \div 8$

9 0.7×0.8

from 7.06

10 $2.4 \div 3$

11 $27.6 \div 4$

12 $40.2 \div 5$

13 $1.2 \div 4$

14 $68.4 \div 9$

15 $8.4 \div 7$

16 $2 \div 5$

Easy

Questions 17–27

17 $34.2 \div 5$

18 $0.4 \div 4$

19 $27.9 \div 3$

20 Name the quadrilateral.



from 6.08

21 $7.4 \div 5$

___ $\div$ ___

= ___

22 $46.71 \div 9$

23 $0.6 \div 2$

24 $5\,000 \div 1\,000$

from 3.03

25 $31 \div 5$

26 $8.9 \div 4$

27 $43.33 \div 7$

Medium

Questions 28–35

28 $3 \div 4$

29 Mia and friends share a \$42.30 bill equally between 5 people. How much each?

30 Order from largest to smallest: 0.05, 0.8, 0.24, 0.861

from 7.01

31 $12.64 \div 4$

32 $6.9 \div 4$

33 Given $141 \div 3 = 47$, find $14.1 \div 3$.

from 7.05

34 Mia and friends share a \$12.52 bill equally between 2 people. How much each?

35 $8.845 \div 5$

Medium

Questions 36–46

- 36 Ali and friends share a \$44.04 bill equally between 6 people. How much each?

37 $31.318 \div 7$

- 38 Work out $6 \times (30 + 9)$ two ways.

from 5.02

39 $6.1 \div 4$

- 40 Write $\frac{56}{100}$ as a decimal.

from 7.02

- 41 Find the mean of 9, 8.4, 5.7, 8.5.

- 42 Chloe and friends share a \$20.22 bill equally between 6 people. How much each?

43 $23.2 \div 10$

from 7.04

44 $24.848 \div 8$

- 45 A rectangle is 4.9 m by 1.4 m. Find its area.

from 7.06

46 $2.6 \div 8$

Challenging

Questions 47–56

- 47 Find the mean of 8.3, 6.9, 8.7, 5.1.

- 48 A 4.4 m plank is cut into 8 equal pieces. How long is each piece?

- 49 The exterior angle is 125° and one interior opposite angle is 70° . Find the other.

from 6.07

- 50 Find the missing number:
 $\square \times 8 = 6.88$

- 51 How many 12s are in 60?

from 3.03

- 52 Find the mean of 1.1, 4.3, 2.4, 5.6.

- 53 A 3.4 m plank is cut into 5 equal pieces. How long is each piece?

54 $10 - 2.48$

from 7.03

- 55 Find the missing number:
 $\square \times 9 = 6.3$

- 56 Given $161 \times 9 = 1\,449$, find 1.61×9 .

from 7.05

Challenging

Questions 57–66

- 57 A 1.7 m plank is cut into 4 equal pieces. How long is each piece?

- 58 Find the missing number:
 $\square \times 9 = 7.92$

- 59 What is 10% of 232?

from 4.11

- 60 Find the mean of 5.6, 9.4, 7.7, 3.9.

- 61 Write $<$ or $>$: $4.24 \square 4.6$

from 7.01

62 $17.744 \div 4$

- 63 A 2.9 m plank is cut into 5 equal pieces. How long is each piece?

64 $3 \div 100$

from 7.04

- 65 Find the missing number:
 $\square \times 9 = 1.71$

66 $(0.8)^2$

from 7.06

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

Make the divisor a whole number: multiply both numbers by 10 (or 100) first.

Watch your teacher.
Copy every step.

$4.8 \div 0.6 = 48 \div 6 = 8.$

E1 $4.8 \div 0.6$

E2 $1.25 \div 0.05$

E3 How many 0.3 L cups can be filled from 2.4 L?

Easy

Questions 1–16

1 $8 \div 0.1$

2 $0.8 \div 0.2$

3 $7 \div 0.1$

4 $7 \div 0.5$

5 $4.8 \div 0.8$

6 $3 \div 0.1$

7 $5 \div 0.1$

8 $1.6 \div 0.2$

9 $30.59 \div 7$

from 7.07

10 $1.5 \div 0.1$

11 $9 \div 0.1$

12 $0.4 \div 0.2$

13 $7.4 \div 0.1$

14 $6 \div 0.2$

15 $4.5 \div 0.9$

16 $2.1 \div 0.1$

Easy

Questions 17–27

17 $2.4 \div 0.3$

18 $5.1 \div 0.1$

19 $3 \div 0.2$

20 Find m^2 when $m = 3$.

from 5.05

21 $23.4 \div 0.9$

$\times \underline{\quad} : \underline{\quad} \div \underline{\quad}$

$= \underline{\quad}$

22 $3.5 \div 0.5$

23 $7.9 \div 0.1$

24 0.7×0.3

from 7.06

25 $4 \div 0.2$

26 $18.4 \div 0.8$

27 $4.5 \div 0.5$

Medium

Questions 28–35

28 $10.4 \div 0.8$

29 $1.14 \div 0.06$

30 Order from largest to smallest: 0.838, 0.6, 0.43, 0.06

from 7.01

31 How many 0.5 L cups can be filled from 2.5 L?

32 $12 \div 0.5$

33 Given $315 \div 9 = 35$, find $31.5 \div 9$.

from 7.05

34 $0.81 \div 0.03$

35 How many 0.4 L cups can be filled from 4.8 L?

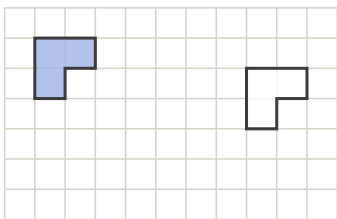
Medium

Questions 36–45

36 $0.84 \div 0.03$

37 How many 0.2 L cups can be filled from 1.6 L?

38 Describe the translation from the shaded shape to the white shape.



from 6.01

39 $7.2 \div 0.3$

40 Write $\frac{92}{100}$ as a decimal.

from 7.02

41 $8.4 \div 1.4$

42 $0.54 \div 0.06$

43 $2.1 \div 3$

from 7.07

44 How many 0.5 L cups can be filled from 3.5 L?

45 A rectangle is 3.1 m by 3.4 m. Find its area.

from 7.06

Challenging

Questions 46–55

46 $13 \div 1.3$

47 Is $3 \div 0.1$ larger or smaller than 3? Explain.

48 Solve $\frac{x}{6} + 5 = 10$.

from 5.12

49 Pencils cost \$0.07 each. How many can you buy with \$1.40?

50 Given $19 \times 3 = 57$, find 0.19×3 .

from 7.05

51 $27.5 \div 2.5$

52 Is $6 \div 0.25$ larger or smaller than 6? Explain.

53 $16 - 1.13$

from 7.03

54 Pencils cost \$0.15 each. How many can you buy with \$1.80?

55 $3.6 \div 4$

from 7.07

Challenging

Questions 56–65

56 Is $8 \div 0.2$ larger or smaller than 8? Explain.

57 Pencils cost \$0.08 each. How many can you buy with \$0.88?

58 What is the order of rotational symmetry of the capital letter H?

from 6.04

59 $4.8 \div 1.6$

60 $33 \div 10$

from 7.04

61 How many 0.4 L cups can be filled from 3.2 L?

62 Is $5 \div 0.5$ larger or smaller than 5? Explain.

63 0.8×4

from 7.06

64 Pencils cost \$0.15 each. How many can you buy with \$0.75?

65 Given $34 \times 8 = 272$, find 3.4×0.8 .

from 7.05

NAME _____ CLASS _____ DATE _____

WE DO

Watch your teacher.
Copy every step.

KEY IDEA

To change a fraction to a decimal, divide the numerator by the denominator. Some decimals go on forever with a repeating pattern: $\frac{1}{3} = 0.333\dots$, written with a dot over the 3.

E1 Write $\frac{3}{8}$ as a decimal.

E2 Write $\frac{2}{3}$ as a decimal.

E3 Write $\frac{5}{11}$ as a decimal.

Easy

Questions 1–9

1 Write $\frac{3}{8}$ as a decimal.

2 Write $\frac{1}{6}$ as a decimal. Use a dot for the repeating digit.

3 Write $0.\dot{3}$ as a fraction.

4 Write $\frac{3}{4}$ as a decimal.

5 Write $\frac{2}{3}$ as a decimal. Use a dot for the repeating digit.

6 Write $0.\dot{1}$ as a fraction.

7 Write $\frac{1}{2}$ as a decimal.

8 Write $\frac{5}{6}$ as a decimal. Use a dot for the repeating digit.

9 Write $\frac{8}{100}$ as a decimal.

from 7.02

Easy

Questions 10–18

10 Write $\frac{1}{9}$ as a decimal. Use a dot for the repeating digit.

11 Write $0.\dot{6}$ as a fraction.

12 Write $\frac{4}{5}$ as a decimal.

13 Classify a triangle with sides 11 cm, 11 cm, 11 cm by its sides.

from 6.05

14 Write $\frac{7}{8}$ as a decimal.

___ ÷ ___
= ___

15 Write $\frac{1}{3}$ as a decimal. Use a dot for the repeating digit.

16 Write $\frac{19}{20}$ as a decimal.

17 $26.67 \div 3$

from 7.07

18 Write $\frac{14}{25}$ as a decimal.

Medium

Questions 19–26

19 Write $\frac{5}{8}$ as a decimal.

20 Write $\frac{1}{3}$ as a recurring decimal.

21 Order from largest to smallest: 0.8, 0.1, 0.579, 0.02

from 7.01

22 Write $2\frac{1}{8}$ as a decimal.

23 Write $\frac{1}{8}$ as a decimal.

24 $0.64 \div 0.08$

from 7.08

25 Write $\frac{5}{11}$ as a recurring decimal.

26 Write $3\frac{2}{5}$ as a decimal.

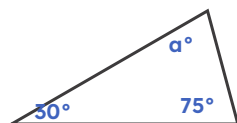
Medium

Questions 27–37

- 27 Write $\frac{1}{11}$ as a recurring decimal.

- 28 Write $5\frac{2}{5}$ as a decimal.

- 29 Find a .



from 6.06

- 30 Write $\frac{6}{25}$ as a decimal.

- 31 $9.6 - 2.5$

from 7.03

- 32 Which is larger: $0.\dot{6}$ or 0.67 ?

- 33 Write $\frac{1}{6}$ as a recurring decimal.

- 34 0.4×0.8

from 7.06

- 35 Write $3\frac{3}{4}$ as a decimal.

- 36 Write 8.37 as a mixed numeral.

from 7.02

- 37 Write $\frac{1}{16}$ as a decimal.

Challenging

Questions 38–47

- 38 Which is larger: $0.\dot{1}$ or 0.11 ?

- 39 Order from smallest to largest: 0.3 , 0.33 , $\frac{3}{8}$, $\frac{1}{3}$

- 40 Solve $\frac{a}{4} = 5$.

from 5.11

- 41 Use a calculator to write $\frac{1}{7}$ as a decimal. How many digits repeat?

- 42 $1.2 \div 4$

from 7.07

- 43 Which is larger: $0.\dot{3}$ or 0.33 ?

- 44 Use a calculator to write $\frac{3}{7}$ as a decimal. How many digits repeat?

- 45 Change 345 mL to L.

from 7.04

- 46 Use a calculator to write $\frac{2}{7}$ as a decimal. How many digits repeat?

- 47 How many 0.4 L cups can be filled from 2.4 L?

from 7.08

Challenging

Questions 48–57

- 48 Write $1\frac{2}{5}$ as a decimal.

- 49 Write $2\frac{2}{5}$ as a decimal.

- 50 Simplify $m \times m$.

from 5.03

- 51 Write $3\frac{1}{4}$ as a decimal.

- 52 Given $18 \times 8 = 144$, find 18×0.8 .

from 7.05

- 53 Write $2\frac{5}{8}$ as a decimal.

- 54 Write $4\frac{2}{5}$ as a decimal.

- 55 8.8×6

from 7.06

- 56 Write $2\frac{1}{4}$ as a decimal.

- 57 Write 0.45 as a fraction in simplest form.

from 7.02

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

To round to 2 decimal places, look at the third decimal place: 5 or more, round up; 4 or less, leave it. Then drop the digits after the second place.

Watch your teacher.
Copy every step.

E1 Round 3.476 to 2 decimal places.

E2 Round 12.849 to 1 decimal place.

E3 Round $\frac{5}{7}$ to 3 decimal places.

Easy

Questions 1–9

1 Round 34.63 to 1 decimal place.

2 Round 28.5 to the nearest whole number.

3 Round 2.391 to 2 decimal places.

4 Round 74.34 to 1 decimal place.

5 Round 70.2 to the nearest whole number.

6 Round 2.139 to 2 decimal places.

7 Round 98.64 to 1 decimal place.

8 Round 24.7 to the nearest whole number.

9 Write $\frac{1}{9}$ as a decimal. Use a dot for the repeating digit.

from 7.09

Easy

Questions 10–18

10 Round 12.4 to the nearest whole number.

11 Round 2.442 to 2 decimal places.

12 Round 88.29 to 1 decimal place.

13 A rhombus has a side of 13 cm. Find its perimeter.

from 6.10

14 Round 1.0135 to 1 decimal place.

look at place ____: ____

$\approx$ ____

15 Round 23.1 to the nearest whole number.

16 Round 5.715 to 2 decimal places.

17 Round 218 to its first digit.

from 3.01

18 Round 24.42 to 1 decimal place.

Medium

Questions 19–26

19 Round 8.2785 to 1 decimal place.

20 Round \$2.766 to the nearest cent.

21 Order from largest to smallest: 0.33, 0.58, 0.5, 0.03

from 7.01

22 Write $\frac{2}{3}$ as a decimal rounded to 2 decimal places.

23 Round 6.5966 to 3 decimal places.

24 $0.36 \div 0.09$

from 7.08

25 Round \$9.404 to the nearest cent.

26 Write $\frac{5}{6}$ as a decimal rounded to 2 decimal places.

Medium

Questions 27–37

- 27 Round \$5.615 to the nearest cent.

- 28 Write $\frac{5}{11}$ as a decimal rounded to 2 decimal places.

- 29 Find a .



from 6.06

- 30 Round 2.1626 to 1 decimal place.

- 31 Write $\frac{62}{100}$ as a decimal.

from 7.02

- 32 \$75 is shared equally by 7 people. How much each, to the nearest cent?

- 33 Round \$6.190 to the nearest cent.

- 34 $9.88 \div 2$

from 7.07

- 35 Write $\frac{1}{6}$ as a decimal rounded to 2 decimal places.

- 36 Write $3\frac{2}{5}$ as a decimal.

from 7.09

- 37 Round 3.0246 to 1 decimal place.

Challenging

Questions 38–46

- 38 \$45 is shared equally by 7 people. How much each, to the nearest cent?

- 39 What is the smallest number with 2 decimal places that rounds to 6.8 (to 1 decimal place)?

- 40 Write 100 g as a fraction of 3 kg.

from 4.12

- 41 Round 65.135 to the nearest whole number, to 1 decimal place and to 2 decimal places.

- 42 Is 7 345 closer to 7 000 or 8 000?

from 3.01

- 43 \$59 is shared equally by 3 people. How much each, to the nearest cent?

- 44 What is the smallest number with 2 decimal places that rounds to 4.8 (to 1 decimal place)?

- 45 $11 - 2.54$

from 7.03

- 46 Round 58.092 to the nearest whole number, to 1 decimal place and to 2 decimal places.

Challenging

Questions 47–56

- 47 What is the smallest number with 2 decimal places that rounds to 8.0 (to 1 decimal place)?

- 48 Round 26.797 to the nearest whole number, to 1 decimal place and to 2 decimal places.

- 49 $8a \times 7c$

from 5.08

- 50 \$65 is shared equally by 6 people. How much each, to the nearest cent?

- 51 $84.1 \div 1\ 000$

from 7.04

- 52 Write $\frac{1}{7}$ as a decimal rounded to 2 decimal places.

- 53 What is the smallest number with 2 decimal places that rounds to 7.1 (to 1 decimal place)?

- 54 $1 \div 5$

from 7.07

- 55 Round 33.245 to the nearest whole number, to 1 decimal place and to 2 decimal places.

- 56 Write $\frac{13}{40}$ as a decimal.

from 7.09

NAME _____ CLASS _____ DATE _____

WE DO

Watch your teacher.
Copy every step.

KEY IDEA

Money has two decimal places: write \$4.50, not \$4.5. Add, subtract, multiply and divide money like any decimal. For cash, round to the nearest 5 cents.

E1 Find the total:
 $\$4.85 + \$12.40 + \$0.95$

E2 Find the change from
\$50 for \$37.65.

E3 Find the cost of 3.5 kg of
apples at \$4.20 per kg.

Easy

Questions 1–9

1 Find the total: $\$15.63 + \1.74

2 Find the change from \$20
for \$17.41.

3 Round \$6.72 to the nearest
5 cents.

4 Find the total: $\$14.03 + \3.11

5 Find the change from \$20
for \$13.12.

6 Round \$8.72 to the nearest
5 cents.

7 Find the total: $\$12.67 + \1.75

8 Find the change from \$10
for \$4.30.

9 $7.5 - 4.8$

from 7.03

Easy

Questions 10–18

10 Find the change from \$10
for \$5.90.

11 Round \$8.85 to the nearest
5 cents.

12 Find the total: $\$6.02 + \7.87

13 Solve in your head: $\frac{k}{3} = 10$

from 5.10

14 Find the cost of 9 pies at
\$9.80 each.

=

15 Find the change from \$20
for \$3.57.

16 Round \$3.69 to the nearest
5 cents.

17 Round 68.3 to the nearest
whole number.

from 7.10

18 Find the total: $\$2.39 + \9.33

Medium

Questions 19–25

19 Find the cost of 4 notebooks
at \$6.85 each.

20 \$8.88, \$5.93 and \$4.96 are
paid with a \$50 note. Find
the change.

21 Order from largest to
smallest: 0.05, 0.1, 0.624, 0.65

from 7.01

22 Find the cost of 4 kg of
grapes at \$4.20 per kg.

23 Find the cost of 9 tickets at
\$6.92 each.

24 Write $\frac{4}{9}$ as a recurring
decimal.

from 7.09

25 \$5.78, \$1.43 and \$7.28 are
paid with a \$50 note. Find
the change.

Medium

Questions 26–35

- 26 \$8.55, \$7.48 and \$8.99 are paid with a \$50 note. Find the change.

- 27 Find the cost of 1.9 kg of grapes at \$1.20 per kg.

- 28 Write an expression: 12 less than b

from 5.04

- 29 Find the cost of 3 notebooks at \$3.46 each.

- 30 Write $\frac{61}{100}$ as a decimal.

from 7.02

- 31 A bill of \$23.18 is split between 6 people. How much does each pay, to the nearest cent? Is the total then exact?

- 32 \$3.27, \$8.08 and \$1.66 are paid with a \$50 note. Find the change.

- 33 $2.8 \div 0.7$

from 7.08

- 34 Find the cost of 2.6 kg of grapes at \$4.20 per kg.

- 35 A \$50 note pays for a \$12.72 meal. Find the change.

from 7.03

Challenging

Questions 36–43

- 36 A bill of \$27.66 is split between 6 people. How much does each pay, to the nearest cent? Is the total then exact?

- 37 Items cost \$4.91 and \$2.75. Paying cash, the total is rounded to the nearest 5 cents. What do you pay?

- 38 11 equal packs hold 319 pencils. How many pencils are in each pack?

from 5.13

- 39 A worker earns \$26.50 an hour. How much is earned for 7.5 hours a day, 5 days a week?

- 40 Round 4.125 to 2 decimal places.

from 7.10

- 41 A bill of \$86.66 is split between 5 people. How much does each pay, to the nearest cent? Is the total then exact?

- 42 Items cost \$4.95 and \$7.74. Paying cash, the total is rounded to the nearest 5 cents. What do you pay?

- 43 Change 877 mL to L.

from 7.04

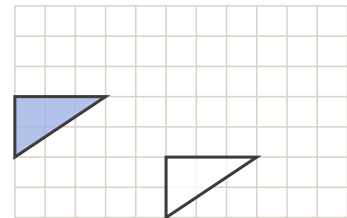
Challenging

Questions 44–51

- 44 Items cost \$1.07 and \$8.82. Paying cash, the total is rounded to the nearest 5 cents. What do you pay?

- 45 A worker earns \$24.50 an hour. How much is earned for 5.5 hours a day, 4 days a week?

- 46 Describe the translation from the shaded shape to the white shape.



from 6.01

- 47 A bill of \$20.72 is split between 4 people. How much does each pay, to the nearest cent? Is the total then exact?

- 48 Given $12 \times 7 = 84$, find 12×0.7 .

from 7.05

- 49 Find the cost of 4 kg of grapes at \$1.20 per kg.

- 50 Items cost \$6.61 and \$4.34. Paying cash, the total is rounded to the nearest 5 cents. What do you pay?

- 51 $8 \div 0.1$

from 7.08